# Pituitary drugs

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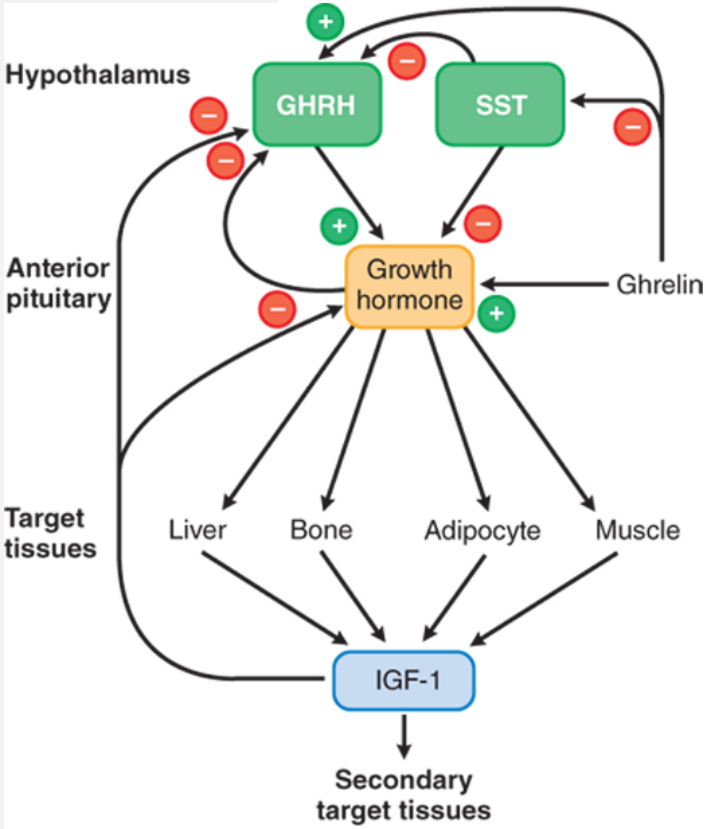
# วัตถุประสงค์

เมื่อสิ้นสุดการบรรยาย นักศึกษาสามารถ

- บอกชื่อยา และอธิบายผลทางเภสัชวิทยาของ pituitary drugs
- บอกการใช้ประโยชน์ทางคลินิก และอาการไม่พึงประสงค์ของ pituitary drugs
  - Hypothalamic-Pituitary-Growth hormone axis
    - Growth hormone and its antagonists
  - Hypothalamic-Pituitary-Gonadal axis
    - Gonadotropin-releasing hormone agonists and its antagonists
    - Gonadotropins (FSH, LH, hCG)
  - Prolactin and dopamine agonists
  - Vasopressin and analog

# Hypothalamic-Pituitary-GH Axis

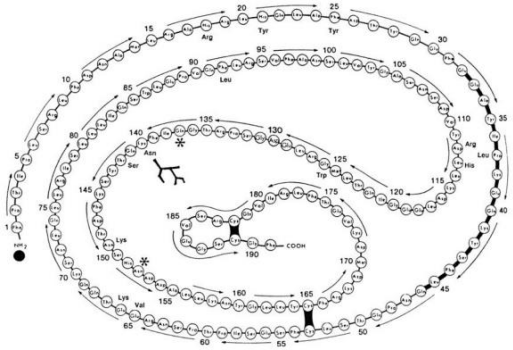
- regulate general processes that promote growth




Source: Laurence L. Brunton, Randa Hilal-Dandan, Björn C. Knollmann: Goodman & Gilman's: The Pharmacological Basis of Therapeutics, Thirteenth Edition: Copyright © McGraw-Hill Education. All rights reserved.

(SST: somatostatin)

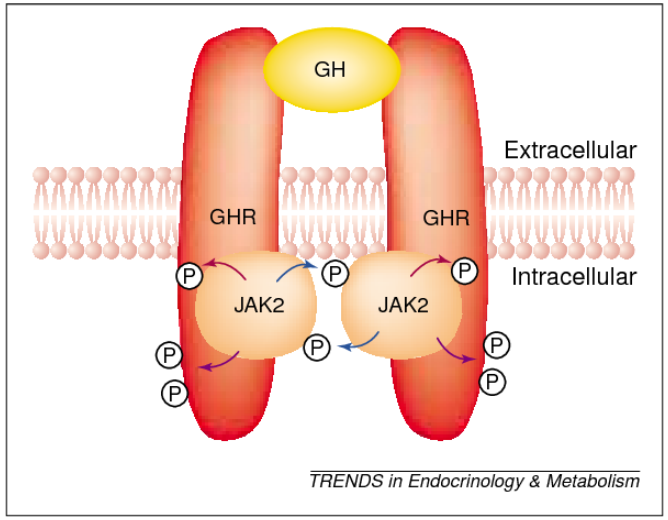
- **Regulation of GH secretion**
  - Somatothropes
  - Stimuli ⇨ GHRH & ghrelin
  - Feedback inhibition: GH, IGF-1, SST
- **Growth hormone (GH)**
  - A 191-amino-acid peptide with two sulfhydryl bridges
  - **Pharmacokinetics**
    - Endogenous GH: half-life 20-25 min
      : cleared by the liver



Human GH structure

# Growth hormone

- **Recombinant human GH (rhGH)**
  - **Somatropin**
  - SC injection
  - half-life 2-3 h




(Trends in Endocrinology & Metabolism,2001;12(6): 252-257.)

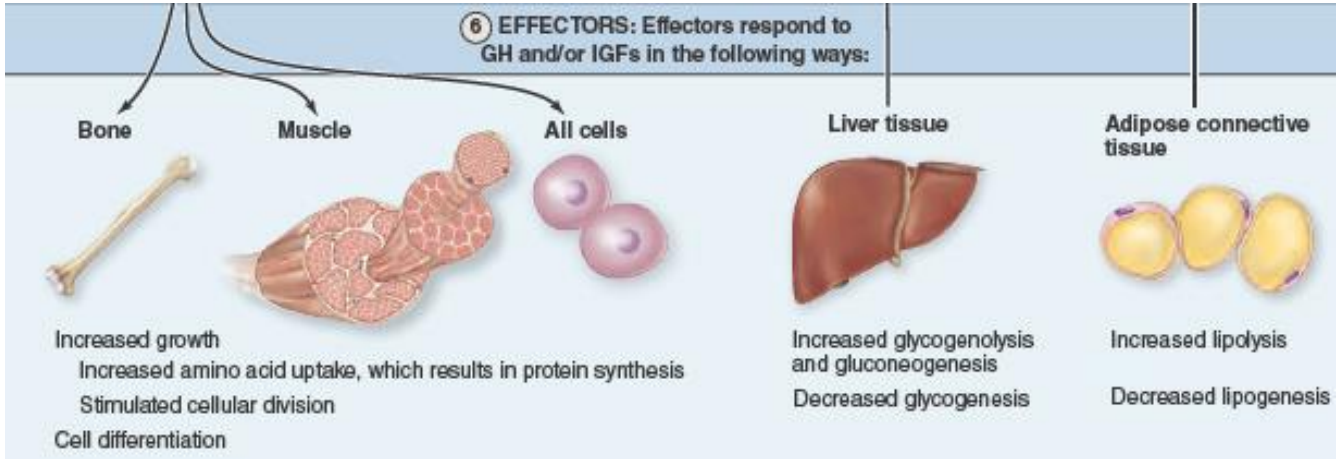
## Mechanism of action

- interact with GH receptor (GHR) (membrane receptor)
- form GH-$[GHR]_2$ ternary complex
- induce a conformational change that activates downstream signaling (JAK2-STAT pathway)

JAK2 (Janus kinase 2)
STAT (signal transducer and activator of transcription)

# Physiological effects of GH

- **Growth promoting effects**:
  - via insulin-like growth factor 1 (IGF-1 or somatomedin C) secreted from hepatocytes
  - stimulate longitudinal bone growth & mineralisation
  - increase protein synthesis in muscle
- **Metabolic effects**
  - anti-insulin actions
  - increase lipolysis




(https://textflowprod-cf.prod.mheducation.com)

# Therapeutic uses of somatropin

## 1. GH deficiency

- failure to secrete GH or to enhance IGF-1 secretion
- **Clinical presentation**
  - **Children**: short stature, often with mild adiposity, delayed bone age, hypoglycemia
  - **Adult**: generalized obesity, reduced muscle mass, asthenia (อ่อนแรง), ↓bone mineral density, dyslipidemia, ↓cardiac output
- **Treatment:** rhGH (somatropin) replacement
- **Therapeutic objective**
  - Children: growth
  - Adult: improved metabolic state, increased lean body mass, sense of well-being

# Therapeutic uses of somatropin

## 2. other conditions associated with short stature

- adequate GH production
- Genetic conditions: Turner syndrome, Prader-Willi syndrome
- Children with idiopathic short stature

## 3. Other uses

- AIDS-associated wasting
- malabsorption associated with the short-bowel syndrome
- anti-aging medicine (unsubstantiated)
- increase muscle mass and athletic performance (drugs banned)

# Adverse effects of somatropin

- **Children:** well tolerate (few side effects)
  : rare effects~ intracranial hypertension, with papilledema, visual changes, headache, nausea, vomiting
  : increased incidence of type 2 DM
- **Adults:** carpal tunnel syndrome, peripheral edema, myalgias, and arthralgias (esp. hands & wrists)
  : occur most frequently in older or obese patients
  : remit with dosage reduction

- **Contraindication:** active malignancy, critical illness, proliferative retinopathy or severe nonproliferative diabetic retinopathy

# Mecasermin

- recombinant human IGF-1 (rhIGF-1) (SC injection)

**Pharmacokinetics**

- completely absorbed
- binding protein: IGFBP-3 and acid labile subunit → prolongs the $t_{1/2}$ of IGF-1 to about 6 h
- Metabolism: liver and kidney

**Therapeutic uses**

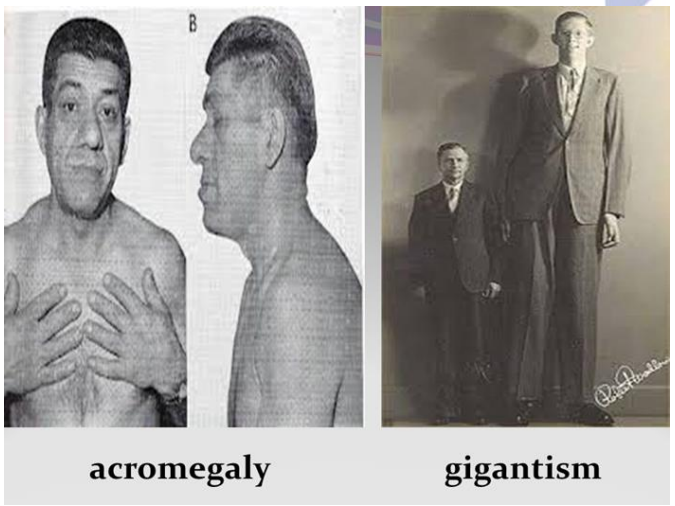
- severe IGF-1 deficiency that is not responsive to exogenous GH
  - mutations in the GH receptor and GH receptor signaling pathway, neutralizing Ab to GH, and IGF-I gene defects

**Adverse effect**

- hypoglycemia (most important), lipohypertrophy
- intracranial hypertension, enlarged tonsils
- asymptomatic elevation of liver enzymes

# GH antagonists

- For treatment of GH secreting pituitary adenoma
  - acromegaly
  - gigantism



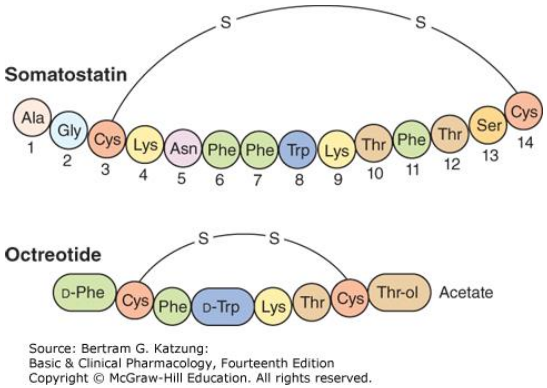

**Treatment**

- Transsphenoidal surgery
- Radiation
- Medications
  - *Somatostatin analogs* → reduce GH production
  - *GH receptor antagonist* → prevents GH from activating GH signaling pathways
  - *Dopamine receptor agonist* → reduce GH production

# Somatostatin analogs

## ❑ Somatostatin (SST)

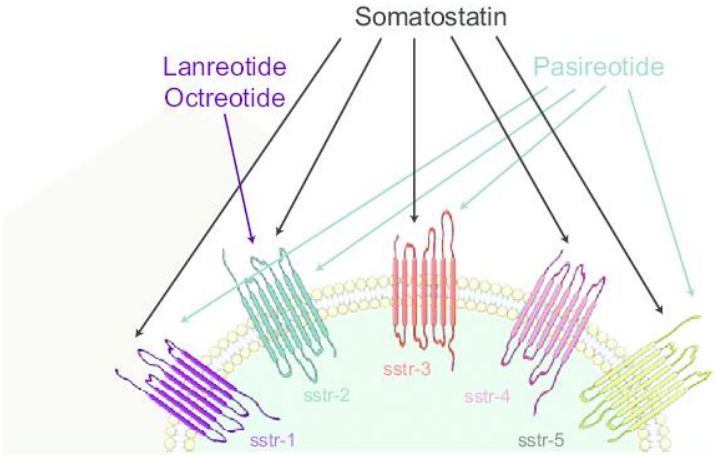
- a 14-amino-acid peptide
- short duration of action ($t_{1/2}$ 1-3 min) → limit its usefulness




Source: Bertram G. Katzung:
Basic & Clinical Pharmacology, Fourteenth Edition
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longer-acting somatostatin analogs
"Octreotide, Lanreotide and Pasireotide"

- binding to SST receptors
- inhibit the release of GH,GI peptide hormones (e.g. insulin, gastrin, CCK)




| Affinity | sstr-1 | sstr-2 | sstr-3 | sstr-4 | sstr-5 |
|---|---|---|---|---|---|
| Somatostatin | 0.9 | 0.2 | 0.6 | 1.5 | 0.3 |
| Octreotide | 280.0 | 0.4 | 7.1 | >1000 | 6.3 |
| Lanreotide | 180 | 0.5 | 14 | 230 | 17 |
| Pasireotide | 9.3 | 1.0 | 1.5 | >1000 | 0.2 |

(Research and Reports in Endocrine Disorders, 2013;3: 31-38.)

# Somatostatin analogs

## ➢ Octreotide

- the most widely used somatostatin analog
- 45 times more potent than STT (inhibit GH release)

**Pharmacokinetics and preparations**

- SC: half-life: 90 min.
  : duration of action 12 h. (q 8 h)
- IM (octreotide LAR): long-acting, slow-release form (q 4 wk)

**Therapeutic uses**

- reduce GH secretion
- certain hormone-secreting tumors
- control of bleeding from esophageal varices

# Somatostatin analogs

## ➢ Octreotide

**Adverse effects**

- GI upset: N/V, abdominal cramps, flatulence, steatorrhea, gallstones,
- injection site pain
- hypothyroidism (inhibit TSH)
- Vit. B12 deficiency (long-term use)

## ➢ Lanreotide

- Lanreotide autogel: long-acting SST analog
- SC injection (q 4 wk)
- for treatment of acromegaly

# Somatostatin analogs

## ➢ Pasireotide

- long-acting SST analog

  **Therapeutic uses**

  - acromegaly
  - Cushing disease

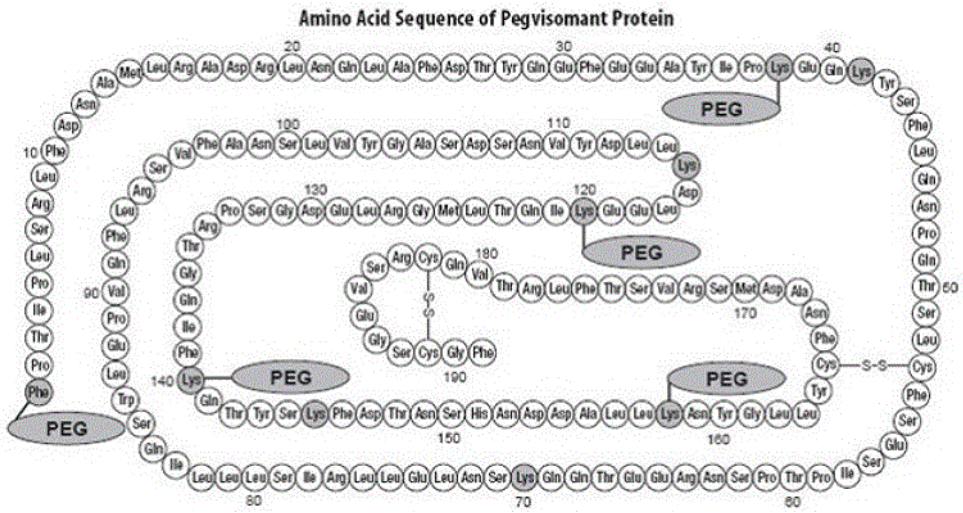
  **Adverse effects**

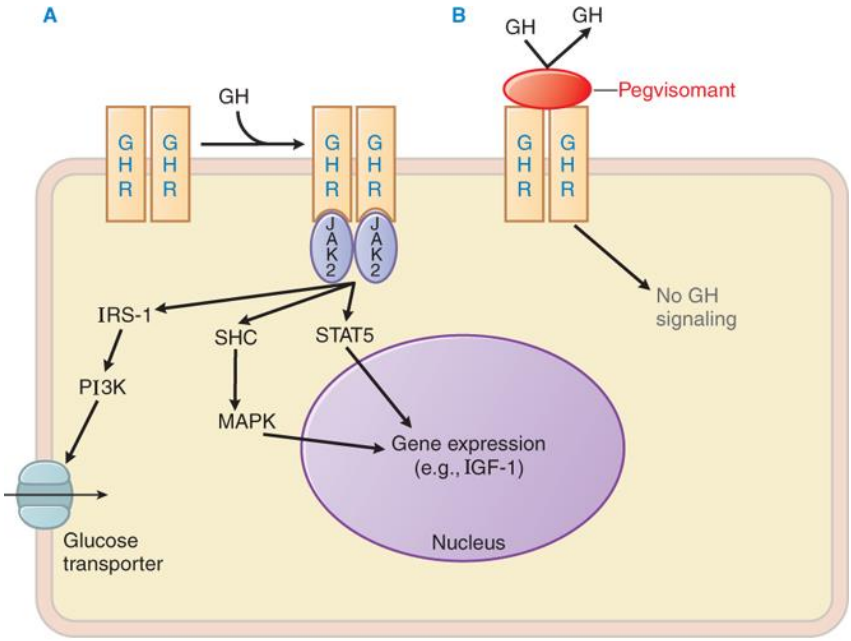
  - Hyperglycemia
    - decrease secretion of insulin, glucagon-like peptide 1 (GLP-1), glucose insulinotropic peptide (GIP)

# GH receptor antagonist

## ➢ Pegvisomant

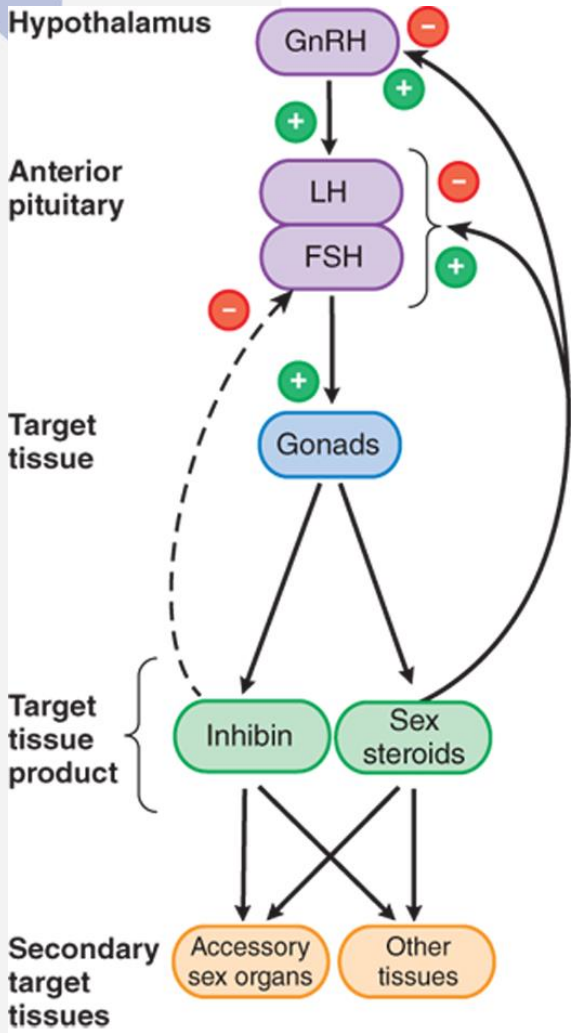
- PEG derivative of a mutant GH (long half-life: 6 days)
- competitive GH receptor antagonist
- used to treat acromegaly (SC)
- bind to GH receptor → blocks the conformational changes required for signal transduction → decrease IGF-1
- Adverse effects: ↑liver enzymes, lipohypertrophy




Source: Randa Hilal-Dandan, Laurence L. Brunton: Goodman and Gilman's Manual of Pharmacology and Therapeutics, 2nd Edition, www.accesspharmacy.com
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# Hypothalamic-Pituitary-Gonadal Axis



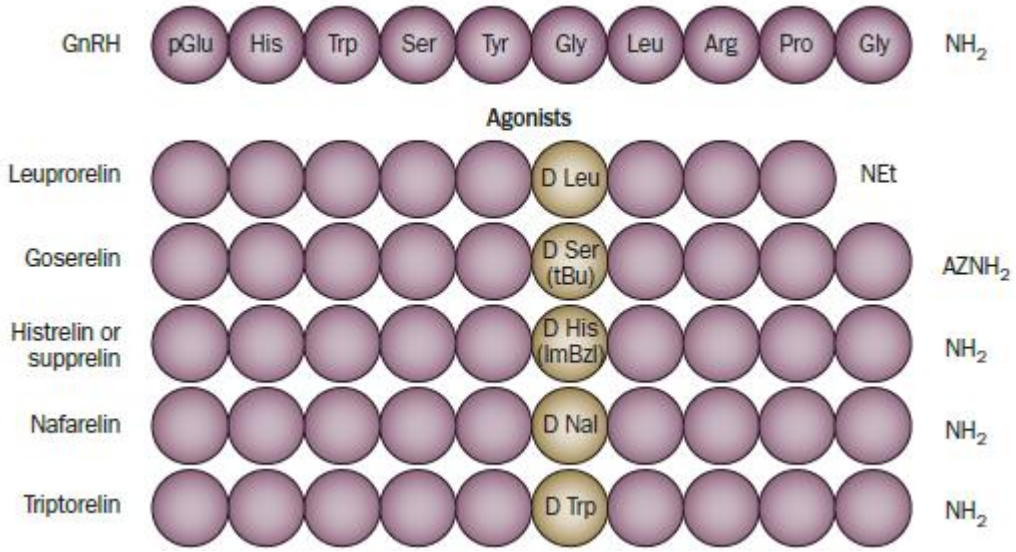

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- Control of reproductive process
  - **Regulation**
    - **Gonadotropin-releasing hormone** (GnRH)
    - **Gonadotropins**:
      - produced by gonadotroph
      - luteinzing hormone (LH)
      - follicle stimulating hormone (FSH)
      - human chorionic gonadotropin (hCG)
      - control sex hormones production
    - Feedback inhibition: sex hormones, inhibins

# Gonadotropin-releasing hormone

## GnRH agonists

- Synthetic human GnRH: Gonadorelin (IV, SC)
- Gonadorelin analogs (SC, IM, implant, intranasal)
  - enhanced potency and prolonged duration of action
  - goserelin, histrelin, leuprolide, nafarelin and triptorelin




(Nature Reviews Endocrinology,2013;9: 451–466.)

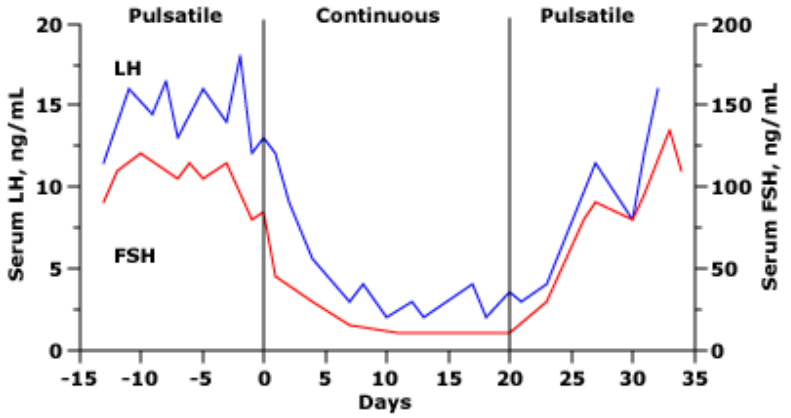
# GnRH agonists action

- **Pulsatile administration** → stimulate LH and FSH secretion
- **Continuous admin./ long acting analog** → biphasic response
  - the first 7-10 days: agonist effect

    → ↑ conc of gonadal hormones in males and females
  - the continued presence of GnRH: inhibitory action

    → ↓ conc of gonadotropins and gonadal steroids

    : due to a combination of receptor down-regulation & changes in the signaling pathways activated by GnRH



# Therapeutic uses of GnRH agonists

## 1. Stimulation of gonadotropin production

- female and male infertility
- uncommon used
- pulsatile IV gonadorelin q 90 min → inconvenience

## 2. Suppression of gonadotropin production

- more common

### 2.1 Prostate cancer

- continuous GnRH agonist + AR antagonist: ↓serum testosterone and prevent tumor flares

### 2.2 Endometriosis

- reduce estrogen & progesterone conc
- duration of treatment: up to 6 months to avoid bone density decrease

# Therapeutic uses of GnRH agonists

## 2. Suppression of gonadotropin production

### 2.3 Uterine fibroids

- reduce fibroid size (temporary)
- for preoperative treatment of uterine leiomyomas

### 2.4 Precocious puberty

- girls: before 7-8 yrs or boys: before 9 yrs
- treatment: continued until age 11 in females & 12 in males

### 2.5 Controlled ovarian stimulation

- for assisted reproduction (in vitro fertilization)
- to suppress an endogenous LH surge that could prematurely trigger ovulation

### 2.6 Other

- advanced breast and ovarian cancer

# GnRH agonists

## Adverse effects

- Gonadorelin: headache, light-headedness, nausea, flushing, local swelling at injection sites
- GnRH analogs (continuous treatment)
  - Menopause-like symptoms in women
    - hot flushes, sweats, headaches, depression, diminished libido, generalized pain, vaginal dryness
  - reduced bone density and osteoporosis (prolong used)
  - Men
    - hot flushes & sweats, edema, gynecomastia, decreased libido, reduced bone density, erectile dysfunction

## Contraindications

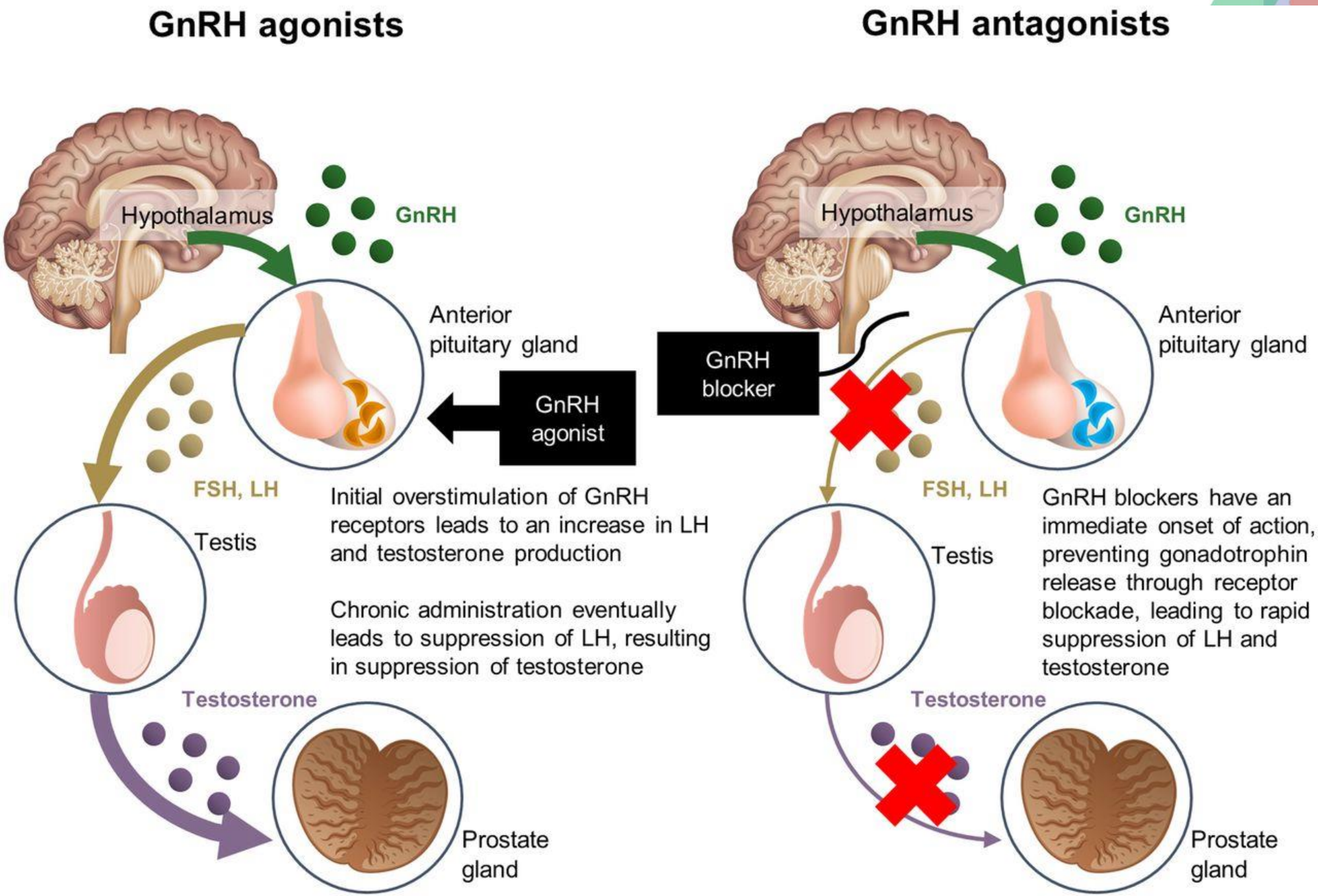
- pregnancy and breast-feeding

# GnRH antagonists

- competitive antagonists of GnRH receptors
- inhibit the secretion of FSH and LH

- **Ganirelix, cetrorelix** (SC)
  - for control ovarian stimulation procedures
    - suppress the LH surge and thus prevent premature ovulation
    - Adverse effects: nausea, headache
- **Degarelix** (SC)
  - for treatment of advanced prostate cancer
    - rapidly reduce gonadotropins and androgen levels
    - avoid the testosterone surge
    - Adverse effects: hot flushes, weight gain, decreased bone mineral density
      : injection-site reactions, ↑ liver enzymes

# Mechanism of action of GnRH agonist and antagonist.




(J Am Coll Cardiol CardioOnc. 2020 Mar, 2 (1) 70-81)

# Gonadotropin preparations

## ❑ Menotropins or human menopausal gonadotropins (hMG)

- first commercial gonadotropin product (IM)
- extracted from urine of postmenopausal women
- contains FSH and LH

## ❑ Follicle-Stimulating Hormone (FSH)

- **Urofollitropin (**uFSH): human FSH (SC)
  - extracted from the urine of postmenopausal women
- **recombinant FSH (rFSH): follitropin alfa, follitropin beta** (SC)
  - shorter half-life than uFSH

## ❑ Luteinizing Hormone (LH)

- **Lutropin alfa:** recombinant human LH (SC)
  - use in combination with follitropin alfa for stimulation of follicular development in infertile women with profound LH deficiency

## ❑ Human Chorionic Gonadotropin

- **Choriogonadotropin alfa** (rhCG): recombinant hCG (IM)

- Half-lives vary by preparation and route of injection from 10 - 40 h.

# Diagnostic Uses of gonadotropins

## 1. Pregnancy Testing

- detect urine hCG using antibodies specific for the unique β subunit of hCG

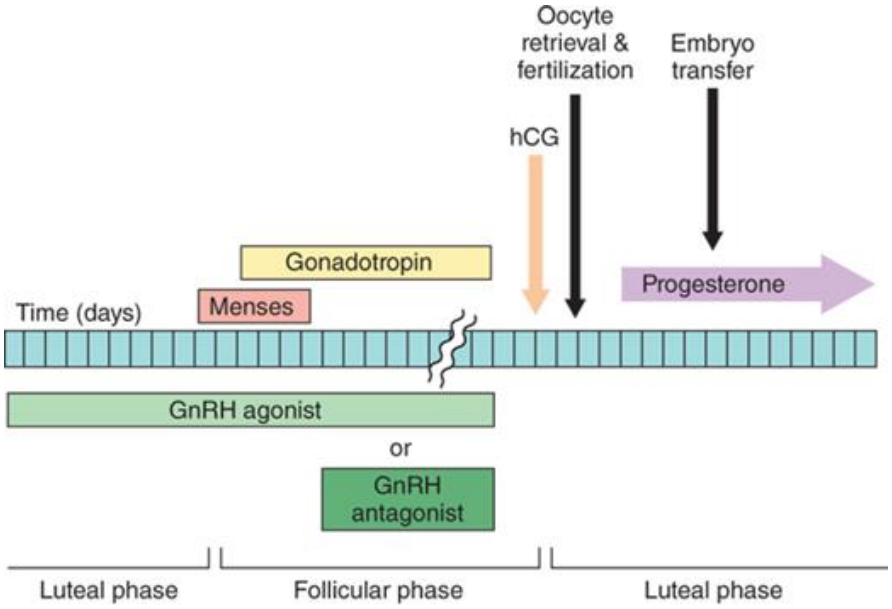
## 2. Timing of Ovulation

- urinary concentrations of LH
- Ovulation occurs about 36 h after the onset of the LH surge
- facilitates the timing of sexual intercourse to optimize the chance of achieving pregnancy

# Therapeutic uses of Gonadotropin

## 1. Ovulation Induction

- induce ovulation in women with anovulation that is secondary to hypogonadotropic hypogonadism, PCOS
- controlled ovarian hyperstimulation in assisted reproductive technology procedures (in vitro fertilization)




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# Therapeutic uses of Gonadotropin

**Adverse effects**

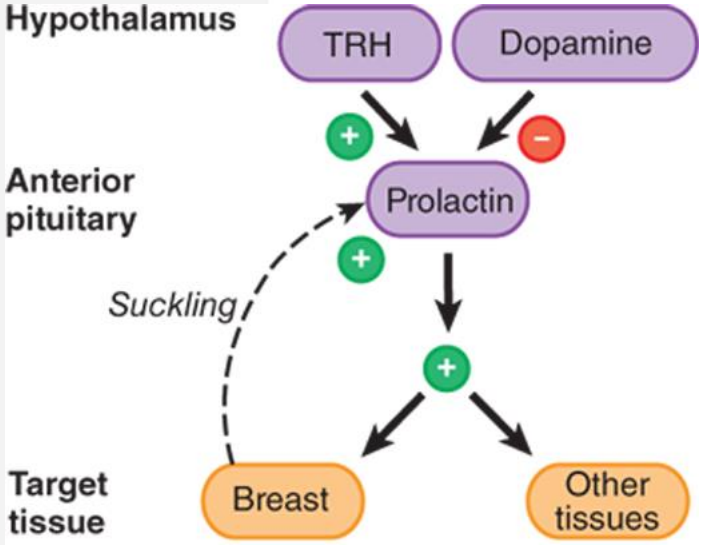
- **ovarian hyperstimulation syndrome (OHSS)**
  - abdominal pain, ascites, pleural effusions, thromboembolism, multiorgan failure
- **multiple pregnancies** (5-10%)
  - ↑ risk of complications: gestational DM, preeclampsia, preterm labor

## 2. Male Infertility

- men with gonadotropin deficiency
- to increase spermatogenesis
- hCG alone (4-6 mo.) or hcG + rFSH
- **Adverse effect**: gynecomastia (most common)

# Prolactin (PRL)

- a 198-amino-acid peptide hormone



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- Target tissue: mammary gland
  : act via PRL receptors
- Effect: mammary development and lactation

## Regulation of secretion

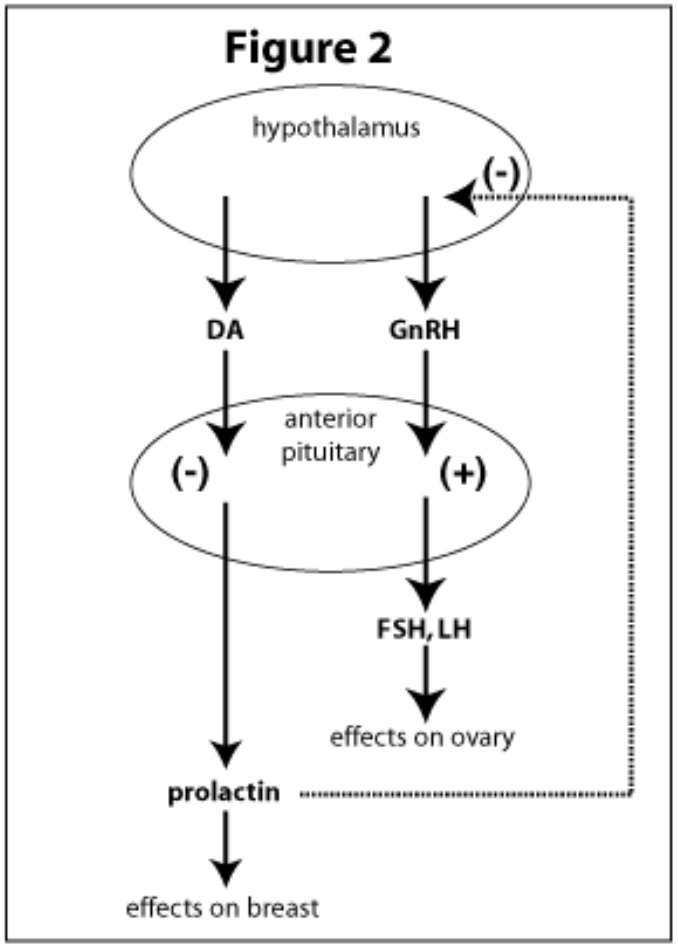
- Major regulator: dopamine (from hypothalamus) interacts with D2 receptor on lactotropes → inhibit PRL release
- Stimulation: suckling, TRH



TRH: thyrotropin-releasing hormone

# Prolactin (PRL)

- **Hyperprolactinemia**
  - **Causes**
    - PRL-secreting adenomas
    - D2 antagonists
    - primary hypothyroidism
  - inhibit GnRH release
  - **Clinical presentations**
    - women: amenorrhea, infertility, galactorrhea
    - men: loss of libido and infertility
  - **Treatment**: dopamine agonists
    → inhibit PRL release from pituitary




**prolactin inhibits GnRH secretion**

(https://courses.washington.edu)

# Dopamine agonists

## ➢ Bromocriptine

- ergot derivatives
- interact with $D_2$ receptors in lactotroph → inhibit PRL secretion
- **Pharmacokinetics**
  - Oral: well absorbed
  - Extensive 1st pass metabolism in liver
  - Short $t_{1/2}$: 7 h
- **Therapeutic uses:**
  - Hyperprolactinemia: reduce pituitary tumor size
    : lower circulating PRL level, restore ovulation
  - Acromegaly: required higher dose
- **Adverse effects:** N/V, headache, postural hypotension, constipation
  : digital vasospasm (high dose)
  : nightmares, insomnia, hallucinations

# Dopamine agonists

## Cabergoline

- ergot derivative with a longer $t_{1/2}$ (~65 h), higher affinity, and greater selectivity for the D2 receptor compared to bromocriptine
- **Adverse effects**
  : nausea (< bromocriptine), hypotension, dizziness
  : valvular heart disease (high dose > 2 mg/week)

## Quinagolide

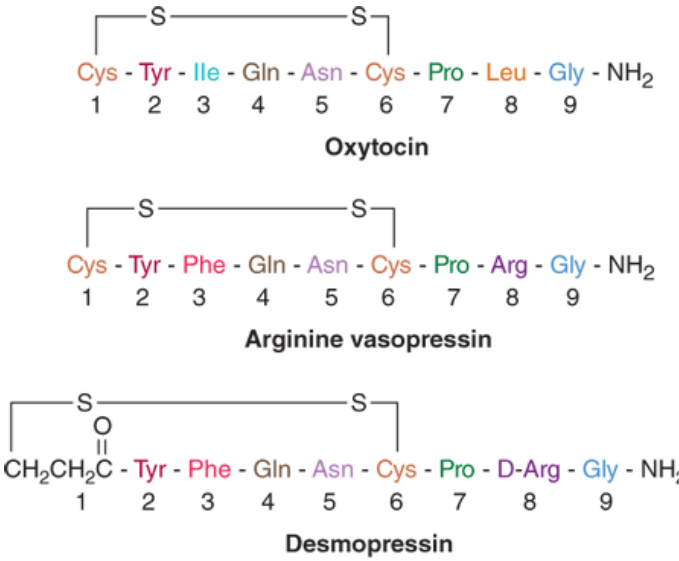
- non-ergot derivative
- high $D_2$ receptor affinity
- $t_{1/2}$ ~22 h
- should not be used when pregnancy is intended
- **Adverse effects**: nausea, headache

# Posterior pituitary hormones

- ❑ Oxytocin (oxytocic drug for induction of labor)
- ❑ Vasopressin (Antidiuretic hormone; ADH)

## Vasopressin (ADH)

- Arginine vasopressin (AVP)
- cyclic nonapeptides hormone
- Vasoconstrictor
  - V1 receptors in vascular smooth muscle cells
- Antidiuretic
  - V2 receptors in renal tubule cells
- ADH deficiency: diabetes insipidus (DI)




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# Vasopressin and its analog

- **Vasopressin**

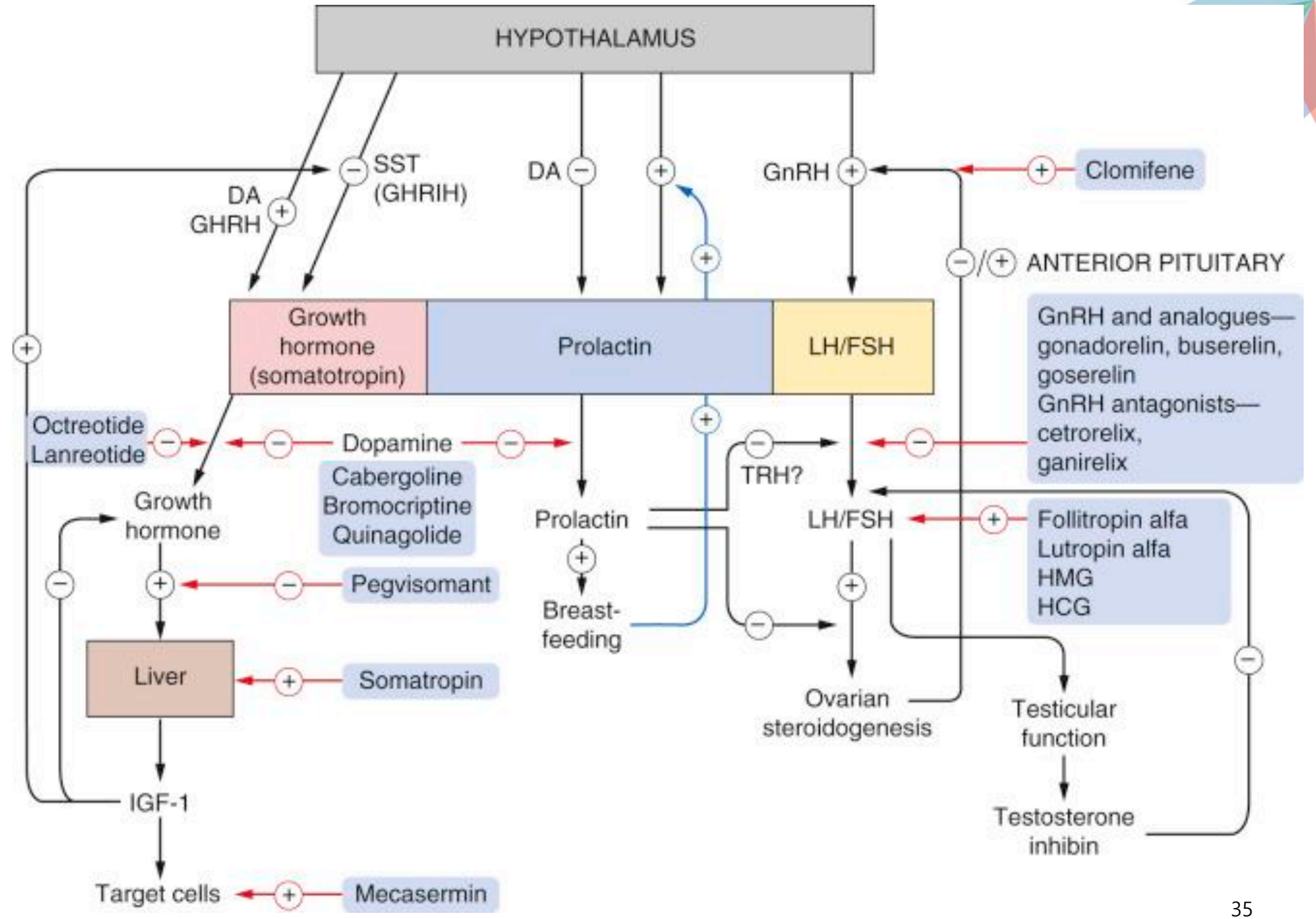
  **Pharmacokinetics**

  - IV or IM injection
  - $t_{1/2}$: 10-15 min
  - Metabolism: renal & liver

- **Vasopressin analog**
  - **Desmopressin, 1-desamino-8-D-arginine vasopressin (DDAVP)**
    - V2–selective agonist
    - antidiuretic-to-vasopressor ratio: 3000 times greater than that for vasopressin
    - $t_{1/2}$: 1.5–2.5 h (injection)
    - Preparations: IV, SC, oral, nasal
    - drug of choice for central diabetes insipidus (CDI)
    - for primary nocturnal enuresis

# Adverse Effects

- Headache, nausea, abdominal cramps, agitation, allergic reactions
- excessive water retention (hyponatremia)
- Vasopressin
  - vasoconstriction
  - extreme caution in patients with coronary artery disease






(Medical Pharmacology and Therapeutics. 2018;43: 491-501.)